# Homework 6

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom

Attaching package: 'scales'


The following object is masked from 'package:purrr':

    discard


The following object is masked from 'package:readr':

    col_factor


New names:
Rows: 147 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (8): ...1, female, age, highstatus, yrsmoke, cigsday, bird, cancer

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

### Data: Association Between Bird-Keeping and Risk of Lung Cancer

A 1972-1981 health survey in The Hague, Netherlands, discovered an association between keeping pet birds and increased risk of lung cancer. To investigate birdkeeping as a risk factor, researchers conducted a case-control study of patients in 1985 at four hospitals in The Hague. They identified 49 cases of lung cancer among patients who were registered with a general practice, who were age 65 or younger, and who had resided in the city since 1965. Each patient (case) with cancer was matched with two control subjects (without cancer) by age and sex. Further details can be found in Holst, Kromhout, and Brand (1988).

Age, sex, and smoking history are all known to be associated with lung cancer incidence. Thus, researchers wished to determine after age, sex, socioeconomic status, and smoking have been controlled for, is an additional risk associated with birdkeeping? The data (Ramsey and Schafer 2002) is found in [`birdkeeping.csv`(data/birdkeeping.csv)].[1]

The paper that this exercise is based upon can be found here and please read it before completing the assignment. (https://www.bmj.com/content/bmj/297/6659/1319.full.pdf)

## Exercise 1

**Part a**

Create a segmented bar chart and appropriate table of proportions showing the relationship between birdkeeping and cancer diagnosis. Summarize the relationship in 1 - 2 sentences.

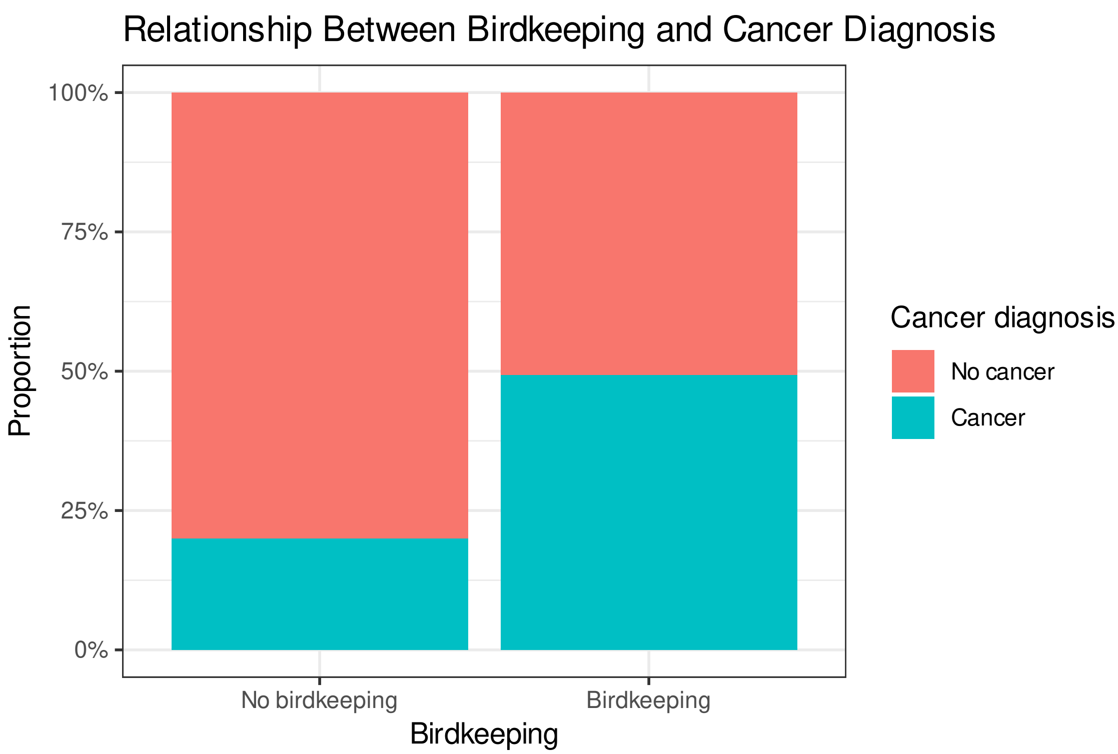
```
# Table of counts
bird_cancer_tab <- table(birdkeeping$bird, birdkeeping$cancer)
# Table of proportions (within birdkeeping group)
bird_cancer_prop <- prop.table(bird_cancer_tab, margin = 1)
# Formatted proportion table
bird_cancer_prop_tbl <- as.data.frame.matrix(round(bird_cancer_prop, 3))
# Rename rows and columns
rownames(bird_cancer_prop_tbl) <- c("No birdkeeping", "Birdkeeping")
colnames(bird_cancer_prop_tbl) <- c("No cancer", "Cancer")

bird_cancer_prop_tbl
```

```
               No cancer Cancer
No birdkeeping     0.800  0.200
```

[1]This problem is adapted from Section 6.8.1, Ex 4.

```
Birdkeeping        0.507  0.493
```

```r
# Segmented bar chart
birdkeeping %>%
  mutate(
    bird = factor(bird, levels = c(0, 1), labels = c("No birdkeeping", "Birdkeeping")),
    cancer = factor(cancer, levels = c(0, 1), labels = c("No cancer", "Cancer"))
  ) %>%
  ggplot(aes(x = bird, fill = cancer)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = percent_format()) +
  labs(
    x = "Birdkeeping",
    y = "Proportion",
    fill = "Cancer diagnosis",
    title = "Relationship Between Birdkeeping and Cancer Diagnosis"
  )
```




Individuals who keep birds have a noticeably higher proportion of cancer diagnoses compared to those who do not keep birds. Conversely, non-birdkeepers have a much larger proportion of individuals without cancer, suggesting a positive association between birdkeeping and cancer risk.

**Part b**

Calculate the unadjusted odds ratio of a lung cancer diagnosis comparing birdkeepers to non-birdkeepers. Interpret this odds ratio in context. (Note: an unadjusted odds ratio is found by not controlling for any other variables.)

```
# Extract counts
a <- bird_cancer_tab[1,1]  # No birdkeeping & No cancer
b <- bird_cancer_tab[1,2]  # No birdkeeping & Cancer
c <- bird_cancer_tab[2,1]  # Birdkeeping & No cancer
d <- bird_cancer_tab[2,2]  # Birdkeeping & Cancer

# Compute odds ratio
OR <- (d * a) / (c * b)

# Create tidy output
or_tbl <- tibble(
  Comparison = "Birdkeeping vs No birdkeeping",
  Odds_Ratio = round(OR, 3)
)

or_tbl
```

```
# A tibble: 1 x 2
  Comparison                    Odds_Ratio
  <chr>                              <dbl>
1 Birdkeeping vs No birdkeeping       3.88
```

The odds of being diagnosed with lung cancer are approximately 3.882 times higher for individuals who keep birds compared to those who do not, based on this unadjusted analysis.

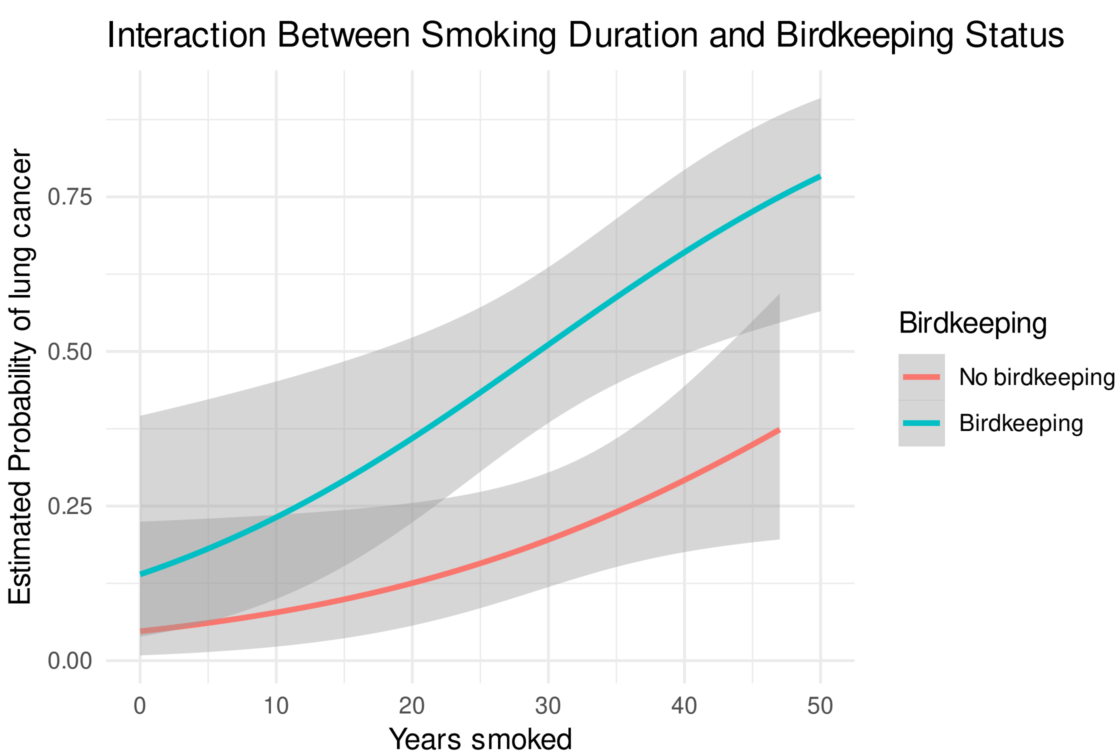
**Part c**

Does there appear to be an interaction between number of years smoked and whether the subject keeps a bird? Demonstrate with an appropriate plot and briefly explain your response.

```
ggplot(birdkeeping, aes(x = yrsmoke, y = cancer, color = factor(bird))) +
  geom_smooth(method = "glm", method.args = list(family = "binomial"), se = TRUE) +
  scale_color_manual(
    values = c("#F8766D", "#00BFC4"),
```

```
    labels = c("No birdkeeping", "Birdkeeping")
  ) +
  labs(
    x = "Years smoked",
    y = "Estimated Probability of lung cancer",
    color = "Birdkeeping",
    title = "Interaction Between Smoking Duration and Birdkeeping Status"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```




The non-parallel curves indicate that the increase in lung cancer risk with years smoked is steeper for birdkeepers than for non-birdkeepers, suggesting a potential interaction between smoking duration and birdkeeping.

**Before answering the next questions, fit logistic regression models in R with cancer as the response and the following sets of explanatory variables:**

- **model1 = age, yrsmoke, cigsday, female, highstatus, bird**
- **model2 = yrsmoke, cigsday, highstatus, bird**

- **model3 = yrsmoke, bird**
- **model4 = yrsmoke, bird, yrsmoke:bird**

```r
# Fit logistic regression models

# Model 1
model1 <- glm(cancer ~ age + yrsmoke + cigsday + female + highstatus + bird,
              data = birdkeeping,
              family = binomial)

# Model 2
model2 <- glm(cancer ~ yrsmoke + cigsday + highstatus + bird,
              data = birdkeeping,
              family = binomial)

# Model 3
model3 <- glm(cancer ~ yrsmoke + bird,
              data = birdkeeping,
              family = binomial)

# Model 4 (with interaction)
model4 <- glm(cancer ~ yrsmoke + bird + yrsmoke:bird,
              data = birdkeeping,
              family = binomial)
```

> **Part d**
>
> Is there evidence that we can remove age and female from our model? Perform an appropriate test comparing model1 to model2; give a test statistic and p-value, and state a conclusion in context.

$H_0 : \beta_{\text{age}} = \beta_{\text{female}} = 0$

$H_A$ : At least one of $\beta_{\text{age}}, \beta_{\text{female}} \neq 0$

We perform a drop in deviance test:

```r
anova(model2, model1, test = "Chisq")
```

```
Analysis of Deviance Table

Model 1: cancer ~ yrsmoke + cigsday + highstatus + bird
Model 2: cancer ~ age + yrsmoke + cigsday + female + highstatus + bird
```

```
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       142     156.72
2       140     154.20  2   2.5257   0.2828
```

The test yields a statistic of $\chi^2 = 2.526$ with $df = 2$ and a p-value of 0.283.

Since the p-value is greater than 0.05, we fail to reject the null hypothesis that the coefficients for age and female are both zero. This suggests that age and sex do not significantly improve the model after accounting for the other variables, and thus it is reasonable to remove them from the model.

> **Part e**
>
> Carefully interpret each of the four model coefficients (including the intercept) in model4 in context.

The fitted model is

$$\log\left(\frac{p}{1-p}\right) = -2.9989 + 0.0528(\text{yrsmoke}) + 1.1795(\text{bird}) + 0.0093(\text{yrsmoke} \times \text{bird}),$$

where $p$ is the probability of lung cancer and bird $= 1$ indicates birdkeeping.

**Intercept:**
The intercept $-2.9989$ is the log-odds of lung cancer for a non-birdkeeper (bird $= 0$) who has smoked 0 years. The corresponding odds are $\exp(-2.9989) \approx 0.050$. Thus, a non-birdkeeper with 0 years of smoking has estimated odds of lung cancer of about 0.050.

**Years smoked (yrsmoke):**
The coefficient 0.0528 represents the change in log-odds of lung cancer for each additional year of smoking among non-birdkeepers. The corresponding odds ratio is $\exp(0.0528) \approx 1.054$. Thus, for non-birdkeepers, each additional year of smoking multiplies the odds of lung cancer by about 1.054, or increases the odds by about 5.4%, holding birdkeeping status fixed.

**Birdkeeping (bird):**
The coefficient 1.1795 represents the difference in log-odds of lung cancer between birdkeepers and non-birdkeepers when yrsmoke $= 0$. The corresponding odds ratio is $\exp(1.1795) \approx 3.253$. Thus, among individuals who have smoked 0 years, birdkeepers have about 3.25 times the odds of lung cancer compared to non-birdkeepers.

**Interaction (yrsmoke $\times$ bird):**
The coefficient 0.0093 indicates how the effect of years smoked differs for birdkeepers relative to non-birdkeepers. The corresponding odds ratio is $\exp(0.0093) \approx 1.009$. Thus, for birdkeepers, each additional year of smoking multiplies the odds of lung cancer by an extra factor of about

1.009 beyond the effect for non-birdkeepers, corresponding to an additional increase of about 0.9% in the odds per year.

For birdkeepers, the total effect of one additional year of smoking is the sum of the smoking coefficient and the interaction coefficient: $0.0528 + 0.0093 = 0.0621$. The corresponding odds ratio is $\exp(0.0621) \approx 1.064$. Thus, for birdkeepers, each additional year of smoking multiplies the odds of lung cancer by about 1.064, or increases the odds by about 6.4%.

> **Part f**
>
> If you replaced yrsmoke everywhere it appears in model4 with a mean-centered version of yrsmoke, tell what would change among these elements: the 4 coefficients, the 4 p-values for coefficients, and the residual deviance.

Mean-centering yrsmoke changes the reference point of the model but does not change the underlying fit.

**Coefficients:** The intercept and the coefficient for bird change because they are now interpreted at $\text{yrsmoke} = \bar{\text{yrsmoke}}$ rather than at $\text{yrsmoke} = 0$. In contrast, the coefficients for yrsmoke and the interaction term $\text{yrsmoke} : \text{bird}$ remain the same because centering does not change the slope or the interaction effect.

**P-values:** The p-values for the intercept and bird change because their interpretations (and thus their standard errors) depend on the reference point of yrsmoke. The p-values for yrsmoke and $\text{yrsmoke} : \text{bird}$ remain unchanged since these terms represent slopes, which are unaffected by centering.

**Residual deviance:** The residual deviance does not change because mean-centering is simply a linear transformation of a predictor and does not affect the fitted probabilities or overall model fit.

> **Part g**
>
> Observe that model3 is a potential final model based on this set of predictor variables. How does the adjusted odds ratio for birdkeeping from model3 compare with the unadjusted odds ratio you found in (b)? Is birdkeeping associated with a significant increase in the odds of developing lung cancer, even after adjusting for other factors?

```r
summary(model3)
```

```
Call:
glm(formula = cancer ~ yrsmoke + bird, family = binomial, data = birdkeeping)
```

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)    
(Intercept) -3.18016    0.63640  -4.997 5.82e-07 ***
yrsmoke      0.05825    0.01685   3.458 0.000544 ***
bird         1.47555    0.39588   3.727 0.000194 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 187.14  on 146  degrees of freedom
Residual deviance: 158.11  on 144  degrees of freedom
AIC: 164.11

Number of Fisher Scoring iterations: 4
```

The unadjusted odds ratio for birdkeeping from part (b) is 3.882. In Model 3, the adjusted odds ratio for birdkeeping is $\exp(1.47555) \approx 4.37$, which is slightly larger than the unadjusted estimate.

This suggests that, after adjusting for years smoked, the association between birdkeeping and lung cancer becomes even stronger. Additionally, the coefficient for birdkeeping in Model 3 is highly statistically significant (p-value $< 0.001$), indicating strong evidence that birdkeeping is associated with an increased odds of developing lung cancer even after accounting for smoking. Thus, birdkeeping appears to have an independent and significant association with lung cancer risk.

> **Part h**
>
> Discuss the scope of inference in this study. Can we generalize our findings beyond the subjects in this study? Can we conclude that birdkeeping causes increased odds of developing lung cancer? Do you have other concerns with this study design or the analysis you carried out?

This study is a case-control study conducted among patients in The Hague, Netherlands, who were registered with general practices and met specific eligibility criteria (e.g., age $\leq 65$ and residency since 1965). Because the sample is not randomly drawn from a broader population, the scope of inference is limited. We can cautiously generalize findings to populations similar to these patients (e.g., similar demographic and geographic contexts), but not to all populations.

Since this is an observational study rather than a randomized experiment, we cannot conclude that birdkeeping causes an increased risk of lung cancer. The results only demonstrate an association. Even though the analysis attempts to adjust for confounders such as age, sex, and smoking history, there may still be residual or unmeasured confounding (e.g., environmental exposures, occupational risks, or socioeconomic differences).

There are also potential sources of bias inherent in the study design. Case-control studies are particularly vulnerable to recall bias, as participants may not accurately report past exposures like smoking or birdkeeping. Selection bias may also arise if controls are not fully representative of the population that produced the cases, despite matching on age and sex. Additionally, measurement error in variables such as years smoked could affect the estimates.

Overall, while the study provides evidence of an association between birdkeeping and lung cancer, the design and potential biases prevent causal conclusions, and the findings should be interpreted with caution.

## Exercise 2

(Ataman and Sarıyer 2021) use ordinal logistic regression to predict patient wait and treatment times in an emergency department (ED). The goal is to identify relevant factors that can be used to inform recommendations for reducing wait and treatment times, thus improving the quality of care in the ED.

The data include daily records for ED arrivals in August 2018 at a public hospital in Izmir, Turkey. The response variable is Wait time, a categorical variable with three levels:

- Patients who wait less than 10 minutes
- Patients whose waiting time is in the range of 10-60 minutes
- Patients who wait more than 60 minutes

> **Part a**
>
> Compare and contrast the proportional odds model with the multinomial logistic regression model. Write your response using 3 - 5 sentences. You can find a brief review of the proportional odds model here: https://library.virginia.edu/data/articles/fitting-and-interpreting-a-proportional-odds-model and https://online.stat.psu.edu/stat504/lesson/8/8.4

The proportional odds model is designed for ordinal response variables and models cumulative log-odds of the form $\log\left(\frac{P(Y\leq k)}{P(Y>k)}\right) = \beta_{0k} - \beta_1 x_1 - \cdots - \beta_p x_p$, where the key feature is that the slope coefficients $\beta_1, \ldots, \beta_p$ are the same across all cutpoints $k$. This implies a common effect of predictors on the odds of being in higher versus lower categories. In contrast, the multinomial logistic regression model is used for nominal outcomes and models category-specific log-odds relative to a baseline, such as $\log\left(\frac{P(Y=k)}{P(Y=1)}\right)$, with different coefficients $\beta_{1k}, \ldots, \beta_{pk}$ for each category. As a result, the proportional odds model is more parsimonious and interpretable, while the multinomial model is more flexible but requires estimating many more parameters. However, the proportional odds model relies on the assumption that predictor effects are constant across categories, which may not always hold.

> **Part b**
>
> Table 5 in the paper contains the output for the wait time and treatment time models. Consider only the model for wait time. Describe the effect of arrival mode (ambulance, walk-in) on the waiting time. Note: walk-in is the baseline in the model. (A link to the paper can be found in the slides).

The p-value for arrival mode is $< 0.001$, indicating that arrival mode is a statistically significant predictor of waiting time after adjusting for the other variables in the model.

The coefficient for arrival mode (ambulance, with walk-in as the baseline) is $-3.398$. In terms of odds, this corresponds to an odds ratio of $\exp(-3.398) \approx 0.033$.

In the proportional odds model, this means that patients arriving by ambulance have substantially lower odds of being in a higher (longer) waiting time category compared to walk-in patients, holding all other variables constant. Specifically, the odds of being in a longer waiting time category decrease by a factor of approximately 0.033 for ambulance arrivals.

Thus, arriving by ambulance is associated with a significantly lower likelihood of longer waiting times (i.e., faster care), after adjusting for other factors in the model.

> **Part c**
>
> Consider output from both the wait time and treatment time models. Use the results from both models to describe the effect of triage level (red = urgent, green = non-urgent) on the wait and treatment times in the ED. Note: red is the baseline level.

For the waiting time model, the coefficient for triage level (green vs. red baseline) is 0.016 with a p-value of 0.153, indicating that triage level is not a statistically significant predictor of waiting time. The small positive coefficient suggests a slight increase in the odds of being in a longer waiting time category for non-urgent (green) patients compared to urgent (red) patients, but this effect is not statistically meaningful.

For the treatment time model, the coefficient for triage level is $-0.950$ with a p-value $< 0.001$, indicating a statistically significant effect. The corresponding odds ratio is $\exp(-0.950) \approx 0.387$, meaning that non-urgent (green) patients have substantially lower odds of being in a longer treatment time category compared to urgent (red) patients.

Overall, triage level does not significantly affect waiting time, but it has a strong and significant effect on treatment time: urgent (red) patients tend to require longer treatment, while non-urgent (green) patients tend to have shorter treatment durations.

## Exercise 3

Ibanez and Roussel (2022) conducted an experiment to understand the impact of watching a nature documentary on pro-environmental behavior. The researchers randomly assigned the 113 participants to watch an video about architecture in NYC (control) or a video about Yellowstone National Park (treatment). As part of the experiment, participants played a game in which they had an opportunity to donate to an environmental organization. The data set is available in nature.csv in the data folder. We will use the following variables:

- donation_binary: 1 - participant donated to environmental organization versus 0 - participant did not donate.
- Age: age in years
- Gender: Participant's reported gender
- Treatment: Urban (T1)" - the control group versus "Nature (T2)" - the treatment group.
- NEP_high: 1 - score of 4 or higher on the New Ecological Paradigm (NEP) versus 0 - score of less than 4.

See the Introduction and Methods sections of Ibanez and Roussel (2022) for more details about the variables and see the class slides regarding the url for the paper.

```
nature <- read_csv("../data/nature.csv", show_col_types = FALSE)
# https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0275806
nature = nature %>% select(c("donation_binary", "Age", "Gender", "Treatment", "nep_high", "Do
nature <- nature %>%
  rename(donation = `Donation (level)`)
summary(nature)
```

```
 donation_binary       Age            Gender         Treatment        
 Min.   :0.0000   Min.   :18.00   Min.   :0.0000   Length:113        
 1st Qu.:1.0000   1st Qu.:20.00   1st Qu.:0.0000   Class :character  
 Median :1.0000   Median :22.00   Median :1.0000   Mode  :character  
 Mean   :0.7788   Mean   :22.58   Mean   :0.5752                     
 3rd Qu.:1.0000   3rd Qu.:24.00   3rd Qu.:1.0000                     
 Max.   :1.0000   Max.   :35.00   Max.   :1.0000                     
    nep_high         donation     
 Min.   :0.0000   Min.   : 0.000  
 1st Qu.:0.0000   1st Qu.: 1.000  
 Median :0.0000   Median : 3.000  
 Mean   :0.4513   Mean   : 3.177  
 3rd Qu.:1.0000   3rd Qu.: 5.000  
 Max.   :1.0000   Max.   :10.000  
```

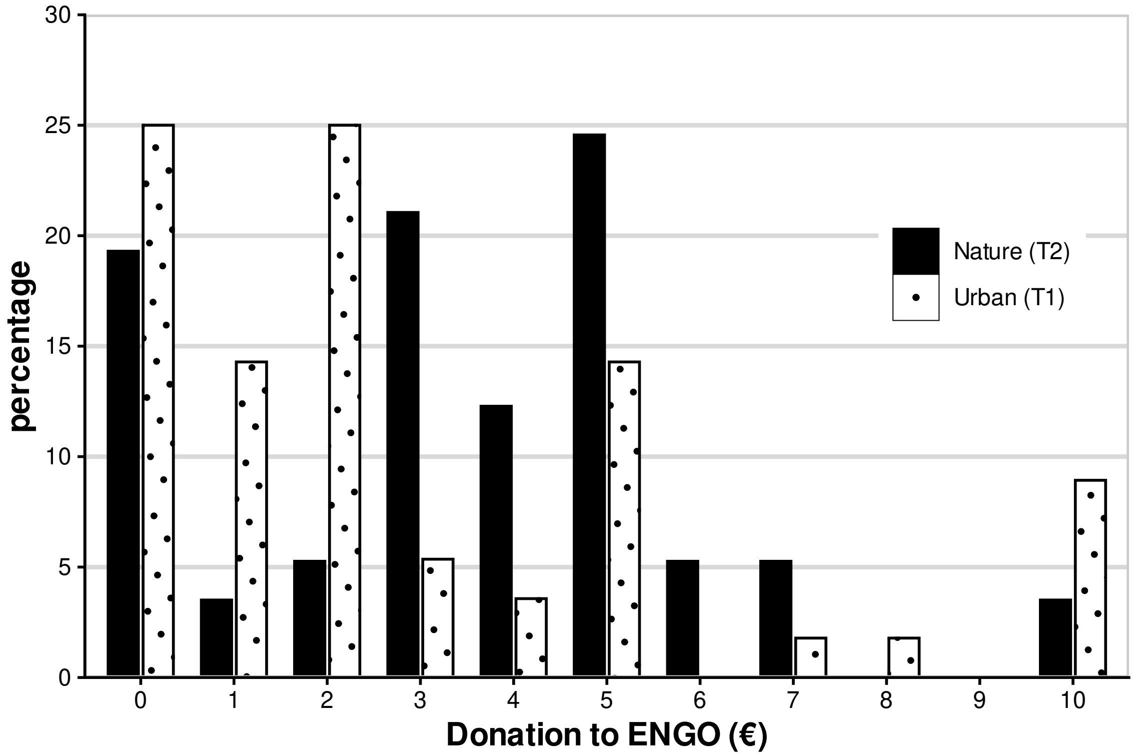
> **Part a**
>
> Figure 2 on pg. 9 of the article visualizes the relationship between donation amount and treatment. Recreate this visualization using your own code. Use the visualization to describe the relationship between donating and the treatment.

```r
install.packages("ggpattern")
```

```
Installing package into '/home/p/pm266/R/x86_64-redhat-linux-gnu-library/4.5'
(as 'lib' is unspecified)
```

```r
library(ggpattern)

nature <- nature %>%
  mutate(
    Treatment = factor(Treatment, levels = c("NATURE (T2)", "URBAN (T1)")),
    donation = factor(donation, levels = 0:10)
  )

nature_plot_data <- nature %>%
  count(Treatment, donation) %>%
  tidyr::complete(Treatment, donation, fill = list(n = 0)) %>%
  group_by(Treatment) %>%
  mutate(percent = 100 * n / sum(n))

ggplot(
  nature_plot_data,
  aes(x = donation, y = percent, fill = Treatment, pattern = Treatment)
) +
  geom_col_pattern(
    position = position_dodge(width = 0.75),
    width = 0.65,
    color = "black",
    pattern_fill = "black",
    pattern_colour = "black",
    pattern_density = 0.15,
    pattern_spacing = 0.04
  ) +
  scale_fill_manual(
    values = c("black", "white"),
    labels = c("Nature (T2)", "Urban (T1)")
```

```r
  ) +
  scale_pattern_manual(
    values = c("none", "circle"),
    labels = c("Nature (T2)", "Urban (T1)")
  ) +
  scale_y_continuous(
    limits = c(0, 30),
    breaks = seq(0, 30, 5),
    expand = c(0, 0)
  ) +
  scale_x_discrete(drop = FALSE) +
  labs(
  x = "Donation to ENGO (€)",
  y = "percentage",
  fill = NULL,
  pattern = NULL
) +
  guides(
    pattern = "none",
    fill = guide_legend(
      override.aes = list(
        pattern = c("none", "circle"),
        fill = c("black", "white"),
        colour = "black"
      )
    )
  ) +
  theme_classic() +
  theme(
    axis.title.x = element_text(face = "bold", size = 12),
    axis.title.y = element_text(face = "bold", size = 12),
    panel.border = element_rect(color = "grey80", fill = NA, linewidth = 0.8),
    panel.grid.major.y = element_line(color = "grey85", linewidth = 0.8),
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    legend.position = c(0.76, 0.70),
    legend.justification = c(0, 1),
    legend.background = element_rect(fill = "white", color = NA)
  )
```




From the visualization, there is a clear difference in donation behavior between the two treatment groups. Participants in the Nature (T2) group tend to donate higher amounts overall, with a larger proportion of observations in the mid-to-high donation levels (e.g., €3–€5 and above). In contrast, participants in the Urban (T1) group are more concentrated at lower donation amounts, including €0, €1, and €2.

This suggests that exposure to the nature documentary is associated with increased pro-environmental behavior, as participants are both more likely to donate and to donate larger amounts. Overall, the distribution for the Nature group is shifted toward higher donation values compared to the Urban group.

> **Part b**
>
> Fit a probit regression model using age, gender, treatment, nep_high and the interaction between nep_high and treatment predict the likelihood of donating. (Note: Your model will be similar (but not exactly the same) as the "Likelihood" model in Table 5 on pg. 11.) Display the model.

> **Part c**
>
> Describe the effect of watching the documentary on the likelihood of donating.

Part d

Based on the model, what is the predicted probability of donating for a 20-year old female in the treatment group with a NEP score of 3?